

Closed-form Second-order Data Shapley for Group Fairness: Down-weighting Biased Training Examples Without Group Labels

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Abstract

A model trained by empirical risk minimization inherits the biases in its data: when a majority group ties an attribute to the label, the model uses that biased attribute and becomes *unfair* to the minority group. We ask whether *closed-form data valuation* can prevent this during training, with no group labels in the model update. We repurpose In-Run Data Shapley (Wang et al. 2024) as an in-training per-sample reweighter, and add an RMS-rescaled second-order gradient–Hessian–gradient cross-term. The cross-term *automatically down-weights biased-majority examples* (on a controlled probe, majority Shapley values are lower than minority, $p = 1.89 \times 10^{-8}$), and a residual ablation shows this comes from a real directional signal, not from smoothing. Across three data-bias regimes — color, texture, and natural images — the method improves worst-group accuracy and **shrinks the majority–minority accuracy gap by 40–47% over ERM**, and it beats a bi-level fairness reweighter (Ren et al. 2018), among methods that use *no group labels* at training (stronger group-label / two-stage methods exist, which we report). On **CelebA** — a structured *image* demographic bias (predict blond hair, sensitive group gender) — it lifts worst-group accuracy by +51 pp and cuts the equalized-odds gap $0.38 \rightarrow 0.16$; on *diffuse* tabular data (Adult, COMPAS) the gain is small. The mechanism needs structured bias.

Introduction

Fairness problems in machine learning often start in the *data*. If, in the training set, a majority group links some attribute to the label — background to object, color to digit, a demographic feature to an outcome — a model trained to minimize average loss will happily use that attribute. Average accuracy looks good, but the model has *learned the bias*, and it fails on the group where the attribute does not hold. This is exactly an unfair model: it works well for the well-represented group and poorly for the under-represented one. In high-stakes uses — lending, hiring, medical triage — this becomes discrimination: the model is least reliable for the group that is already least represented. Resisting data bias *during* training, rather than patching the model afterward, is a core fairness goal.

The strongest fairness fixes assume extra supervision: group labels at training time (GroupDRO, Sagawa et al. 2020), a second training stage (Liu et al. 2021), or a bi-level meta-loop (Ren et al. 2018). We ask a different question:

can a closed-form, single-pass signal — computed during ordinary training, with no group labels in the update — tell us which examples are teaching the bias, so we can down-weight them?

Our starting point is In-Run Data Shapley (Wang et al. 2024), which gives a closed-form estimate of how much each training example helps a small, balanced reference set, *during* the run. We turn this valuation into a per-sample weight, and study what its *second-order* (curvature) term adds. We find the curvature term is the part that targets the bias.

This connects two course themes — data valuation and group fairness — by turning a *valuation* estimator into a *bias-mitigation* signal inside the optimizer. Our contributions:

- A closed-form In-Run Data Shapley reweighter for **fair training**: it down-weights examples that teach data bias, using no group labels in the update.
- A second-order gradient–Hessian–gradient cross-term with **RMS rescaling**, required for stability under strong bias.
- A **residual ablation** proving the bias-correcting signal is a real, sample-specific direction (the orthogonal residual), not smoothing.
- Across **three data-bias regimes** the method raises worst-group accuracy and shrinks the majority–minority gap by 40–47% over ERM.
- An **honest scope**: a controlled probe of the mechanism, a clean boundary on diffuse tabular demographic bias, and plainly stronger baselines.

Related Work

Group fairness and worst-group accuracy. A common operational notion of group fairness is high *worst-group* accuracy and a small gap between groups (Sagawa et al. 2020). GroupDRO uses group labels; JTT (Liu et al. 2021) and IRM (Arjovsky et al. 2019) relax that need. These are strong but heavier; we offer a closed-form alternative and report the gap honestly.

Fairness by reweighting. Learning to reweight (Ren et al. 2018) and FORML (Yan, Seto, and Apostoloff 2022) solve a bi-level meta problem; ARL (Lahoti et al. 2020) reweights

Algorithm 1 Fairds reweighting step (second order)

Require: minibatch $\mathcal{B}$, balanced anchor $\mathcal{D}_{\text{val}}$, params θ_t , EMA buffer e , scalars α, τ

- 1: $g_i \leftarrow \nabla_{\theta} L(z_i; \theta_t)$ for $z_i \in \mathcal{B}$ // *vmap, single graph*
- 2: $g_{\text{val}} \leftarrow \nabla_{\theta} L(\mathcal{D}_{\text{val}}; \theta_t)$; $Hg \leftarrow \text{Pearlmutter}(\theta_t, g_{\text{val}})$
- 3: $\phi_i^{(1)} \leftarrow \langle g_i, g_{\text{val}} \rangle$; $c_i \leftarrow \langle g_i, Hg \rangle$
- 4: $\tilde{\phi}_i \leftarrow \phi_i^{(1)} - \alpha \frac{\text{RMS}(\phi^{(1)})}{\text{RMS}(c)} c_i$ // *RMS rescale*
- 5: $e_i \leftarrow \text{EMA}(e_i, \tilde{\phi}_i)$; $w \leftarrow B \cdot \text{softmax}(e_B/\tau)$, clipped
- 6: $\theta_{t+1} \leftarrow \theta_t - \eta \sum_i w_i \nabla_{\theta} L(z_i; \theta_t)$

adversarially without demographics. Our reweighter is closed-form and single-pass — no inner loop.

Data valuation. Data Shapley (Ghorbani and Zou 2019) values examples by marginal contribution but needs many retrains; In-Run Data Shapley (Wang et al. 2024) makes it closed-form in one run. FairShap (Arnaiz-Rodriguez and Oliver 2023) uses Shapley values for *fair data valuation*; we instead inject a second-order Shapley signal into the optimization trajectory to mitigate bias as it is being learned.

Method: Fairds

Setup. At step t with parameters θ_t , a minibatch $\mathcal{B} = \{z_i\}_{i=1}^B$ and a small *group-balanced anchor* $\mathcal{D}_{\text{val}}$ (50:50; group labels are used only to build the anchor once, never in the model update). Let $g_i = \nabla_{\theta} L(z_i; \theta_t)$ and $g_{\text{val}} = \nabla_{\theta} L(\mathcal{D}_{\text{val}}; \theta_t)$.

First order. $\phi_i^{(1)} = \langle g_i, g_{\text{val}} \rangle$ (Wang et al. 2024) measures how much z_i reduces loss on the balanced anchor in one SGD step. Per-sample gradients use `torch.func.vmap` in a single graph — closed form, no bi-level loop.

Second order with RMS rescaling. We add the gradient–Hessian–gradient cross-term $c_i = \langle g_i, H_{\text{val}} g_{\text{val}} \rangle$, with $H_{\text{val}} g_{\text{val}}$ from the Pearlmutter trick (Pearlmutter 1994) (one extra backward pass, no explicit Hessian). Since the spectrum of H_{val} varies by orders of magnitude, a fixed α is unusable; we **rescale the cross-term to the same RMS magnitude as the first-order term** each minibatch:

$$\tilde{\phi}_i^{(2)} = \phi_i^{(1)} - \alpha \frac{\text{RMS}(\phi^{(1)})}{\text{RMS}(c)} c_i. \quad (1)$$

Without this rescaling, the method collapses to chance under strong bias. Examples with low $\tilde{\phi}$ — those that help the biased majority but not the balanced anchor — are down-weighted.

Stabilizers. Two choices make the reweighter robust. The EMA buffer averages each example’s score over the steps where it appears, so one noisy minibatch cannot dominate its weight. The per-batch softmax weights are clipped at quantile bounds, so no example collapses to zero or explodes — important under strong bias, where a few majority examples would otherwise be driven to negligible weight.

Why the cross-term targets bias. g_{val} is nearly the top eigenvector of H_{val} (alignment 0.89–0.99), so the normalized cross-term c_n is almost parallel to $\phi^{(1)}$. Decomposing $c_n = \beta \phi^{(1)} + r$ separates a *parallel* part (a global shrinkage, like smoothing) from an *orthogonal residual* r . Whether r is a real bias signal is tested in Section .

Experiments

We measure group fairness three ways: **worst-group accuracy** (higher is fairer; the max–min notion), the **group accuracy gap** between the biased-majority and the minority group (lower is fairer), and, for tabular data, the **demographic-parity (DP)** and **equalized-odds (EO)** gaps. We compare Vanilla SGD; our Fairds-1/2 and the residual variant; a corrected one-step bi-level reweighter (Ren et al. 2018, a FORML proxy); JTT (Liu et al. 2021); and GroupDRO (Sagawa et al. 2020). Vanilla, Fairds, Ren2018 and JTT use *no group labels* in the update; GroupDRO uses them.

Setup

Three from-scratch image benchmarks share one bias structure and differ only in the biased attribute. **Colored MNIST** (Arjovsky et al. 2019): digit ≥ 5 with a spurious *color* attribute; 5,000 images; two-conv CNN. **Corrupted-CIFAR**: car vs. truck with the *corruption type* (blur vs. noise) as the biased attribute; 5,000 images; two-conv CNN. **Spurious-STL10**: the same on 96×96 natural photos; $\sim 2,600$ images; three-conv CNN. In each, the majority group (90% of train) ties the attribute to the label with probability 0.9, the minority 0.5, and the test set flips it to 0.1 — so a model that learned the bias fails on the minority. We also use two tabular **demographic** benchmarks: **Adult** (sex) and **COMPAS** (race). Each reweighting method uses a 200-sample balanced anchor for its inner computation only; a disjoint held-out set is used for model selection; the test set is never seen in training. We fix $\alpha=0.5$, EMA the scores, and use softmax temperature τ with a per-regime weight scale. All numbers are means over 10 seeds (20 for the ablation); significance is the seed-paired Wilcoxon test.

Group fairness across bias regimes

Figures 1 and 2 tell one story. Among no-group-label methods, the second-order residual variant is the fairest in every regime: it gives the highest worst-group accuracy and the smallest majority–minority gap. It **shrinks the accuracy gap by 40–47% over ERM** (0.223→0.118 on MNIST, 0.703→0.430 on CIFAR, 0.950→0.568 on STL10). On the hardest, most natural regime (STL10), it beats ERM by +28.1 pp worst-group ($p=0.002$), the bi-level reweighter by +17.2 pp ($p=0.002$), and the first-order term by +12.9 pp ($p=0.010$). The same second-order isolation reproduces on MNIST (+4.7 pp, $p=0.033$) and CIFAR (+4.9 pp, $p=0.042$). The gain widens as the bias gets harder, and the ranking is stable: one mechanism, three biased attributes. JTT and GroupDRO are fairer still, but use more supervision (a second stage or group labels). In fairness terms, the residual variant moves the *no-label frontier*: among methods

Group fairness across three data-bias regimes: the 2nd-order Shapley reweighter protects the under-represented group (grey/blue = no group label; red = uses group label or 2-stage)

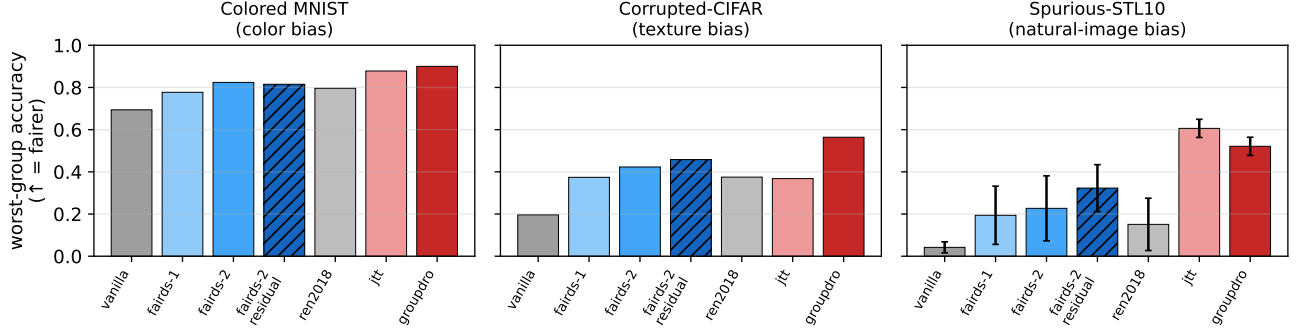


Figure 1: **Group fairness across three data-bias regimes.** Worst-group accuracy (higher = fairer). Grey/blue bars use *no* group label at training; red bars use a group label or a two-stage procedure. The second-order residual variant (hatched) protects the under-represented group better than every no-group-label baseline, in all three regimes. STL10 error bars are ± 1 std (10 seeds).

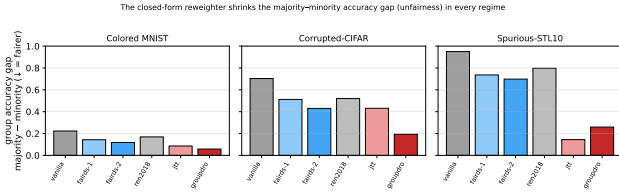


Figure 2: **Unfairness (group accuracy gap), lower is fairer.** The closed-form reweighter shrinks the majority–minority gap below ERM, Fairds-1, and the bi-level baseline in all three regimes.

that never see a group label in the update, it attains the best worst-group accuracy (max–min fairness) and the smallest group gap, and the gain grows with the strength of the bias — exactly where unfairness is most severe and a fix matters most.

The gap closes by protecting the minority. A fairness method should *help* the under-represented group, not equalize by hurting everyone. Our reweighter does the former. On STL10, minority-group (spurious-flipped) accuracy rises $0.04 \rightarrow 0.32$ while the majority dips only $0.99 \rightarrow 0.89$, and *overall* accuracy still climbs $0.14 \rightarrow 0.38$; on CIFAR, minority rises $0.20 \rightarrow 0.46$ with only a small majority dip. The closed-form Shapley signal reallocates learning effort toward the examples the biased majority would otherwise drown out — exactly the group that an unfair model fails. This is why both fairness measures, worst-group accuracy (Figure 1) and the majority–minority gap (Figure 2), improve together.

Why it works: the bias signal is the residual

Is the second-order term a real bias signal or just smoothing? We build five arms that share $\phi^{(1)}$ but treat the residual r differently: *phi1* (first order), *parallel* (shrinkage only), *residual-real* ($\phi^{(1)} - \alpha r$), *residual-shuffle* (r permuted within

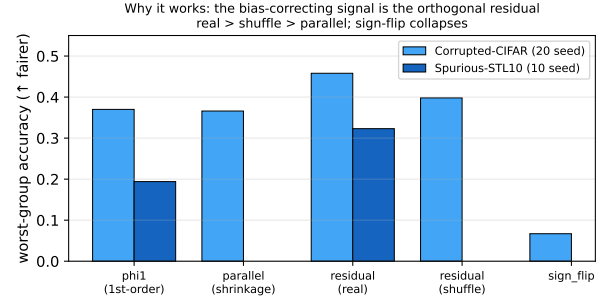


Figure 3: **Mechanism.** The orthogonal residual of the cross-term carries the bias-correcting signal: *real* > *shuffle* > *parallel*; *sign-flip* collapses.

class $\times \phi^{(1)}$ bins), and *sign-flip*. The order is unambiguous: **real** > **shuffle** > **parallel**, and *sign-flip* collapses (Figure 3). On CIFAR, *real* beats *shuffle* by +5.9 pp ($p=0.0015$, 20 seeds) and the shrinkage arm by +9.2 pp. If r were noise, shuffling it within matched bins would not matter — but it does. So the bias-correcting signal is a real, sample-specific direction, not smoothing.

Controlled probe. On a controlled spurious-feature MLP (one strongly label-correlated feature in the majority group), Fairds-2 assigns *lower* Shapley values to majority than minority examples (Mann–Whitney $U(\text{maj} < \text{min})$, $p=1.89 \times 10^{-8}$ at ratio 0.9), and $\Delta\phi$ falls as imbalance grows. This is the bias-mitigation mechanism, seen directly: the cross-term penalizes majority gradients aligned with the validation curvature.

Demographic fairness: structured vs diffuse

Does this help *real* demographic bias? It depends on whether the bias is structured. Table 1 tests two kinds. On **CelebA** — predict *Blond_Hair* with gender (Male) as the sensitive group; blond strongly correlates with female,

data	method	worst $\uparrow$	DP $\downarrow$	EO $\downarrow$	acc
CelebA (image)	Vanilla	0.20	0.237	0.381	0.773
	<i>Fairds-2</i>	0.72	0.213	0.158	0.856
	Ren2018	0.79	0.209	0.129	0.856
Adult (tab.)	Vanilla	0.80	0.196	0.117	0.837
	<i>Fairds-2</i>	0.80	0.189	0.127	0.835
	Ren2018	0.78	0.175	0.102	0.819
COMPAS (tab.)	Vanilla	0.66	0.240	0.226	0.667
	<i>Fairds-2</i>	0.66	0.239	0.244	0.671
	Ren2018	0.67	0.231	0.241	0.681

Table 1: **Demographic fairness.** On **CelebA** (structured image bias) Fairds lifts worst-group +51 pp and cuts EO sharply; on *diffuse* tabular data (Adult, COMPAS) it barely moves worst-group, DP, or EO — the mechanism needs structured bias.

so blond-male is the tiny minority — the bias is structured (one dominant attribute in image space), and Fairds works strongly: worst-group (blond-male) accuracy rises 0.20 $\rightarrow$ 0.72 (+51 pp, $p=0.002$, 10 seeds), the equalized-odds gap falls 0.38 $\rightarrow$ 0.16, and overall accuracy *improves*. It nearly matches the bi-level baseline (Ren2018 worst 0.79). On **Adult** and **COMPAS** — tabular bias spread over many weakly informative features — Fairds keeps accuracy but barely moves DP/EO, and the bi-level method is fairer. The contrast is the point: the closed-form Shapley signal needs a *structured* bias to fire. It is a fairness tool for structured demographic bias — even on real faces — not a general tabular debiaser.

Discussion and Conclusion

Closed-form data valuation can make training fairer, *within a characterized regime*. The second-order Shapley cross-term down-weights the biased-majority examples through its orthogonal residual — a real, per-sample direction — and this raises worst-group accuracy and shrinks the majority-minority gap across color, texture, and natural-image bias. The effect is strong when the bias is structured (one dominant attribute) and weak when it is diffuse, as in tabular demographic data, where a bi-level reweighter is the better tool. A further property is the *fairness-accuracy trade-off*: Fairds improves group fairness while *preserving* overall accuracy, whereas the bi-level reweighter trades accuracy for fairness (Table 1). Closed-form valuation thus gives a distinct operating point — moving toward fairness at almost no accuracy cost — which is attractive when accuracy must be kept and group labels are unavailable. The method is not the fairest overall: two-stage (JTT) and group-label (Group-DRO) methods are stronger, and the second-order variant costs $\sim 7.7\times$ vanilla wall-clock (the first-order variant, $4.3\times$, is cheaper than the bi-level baseline). Its value is conceptual: a single-pass, closed-form valuation whose curvature term provably carries a usable bias-mitigation signal. **Limitations:** (i) not state-of-the-art fairness; (ii) second-order overhead; (iii) each image benchmark isolates one biased attribute; (iv) the diffuse-bias failure is diagnosed empirically,

not proven; (v) the anchor needs group balance at construction, though not in the update. Reducing the overhead and extending the mechanism to diffuse bias are the natural next steps.

Broader Impact

Our goal is fairness, and the method is meant to *reduce*, not certify, unfairness. Two points matter for responsible use. First, the gains are real but bounded: the reweighter raises worst-group accuracy and shrinks the group gap when the bias is structured, yet barely moves demographic-parity or equalized-odds on diffuse tabular data. It should therefore not be deployed as a sole fairness guarantee, and group-label methods remain preferable when that supervision is available. Second, the balanced anchor uses group labels at construction time (though never in the model update); practitioners must check that those labels are themselves trustworthy and that the chosen groups are the ones that matter. Used within its scope — a cheap, label-free training-time signal that keeps structured bias out of the model — the method complements rather than replaces existing fairness tools.

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